

Risk Signal Engine

From breaking news and social posts to explainable risk signals and a portfolio stress test

S&P Global × CRISIL · Code to Connect 2026 · Module B: Strategic Portfolio Stress Testing

1,325

real captured headlines and posts

1–10

explainable impact score

seconds

from headline to portfolio view

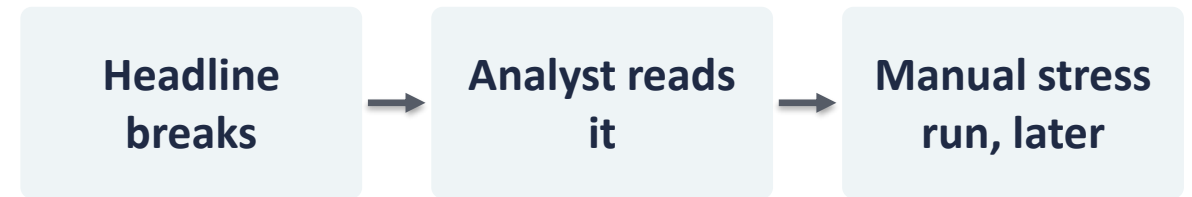
Apeksha Jain · ABV-IIITM Gwalior · individual submission · decision support, not investment advice

Problem and approach

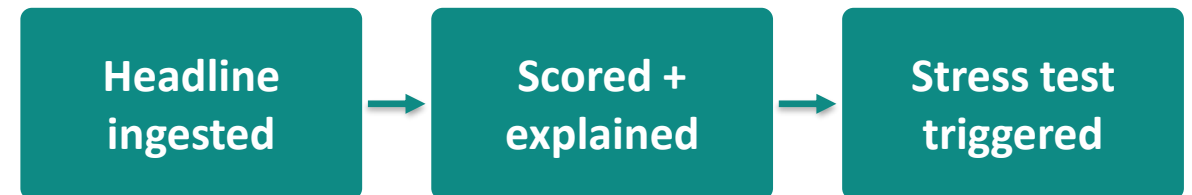
Risk shows up in text before it shows up in data

- Downgrades, probes, sanctions and rate moves break as headlines and posts
- Credit analysts and risk teams triage them by hand: slow, noisy, duplicated
- Nothing links a headline to the book's exposure until someone runs numbers
- **Approach:** an NLP risk engine scores every item (sentiment, event, impact) and an event-driven stress engine reprices the book

Today



With the engine



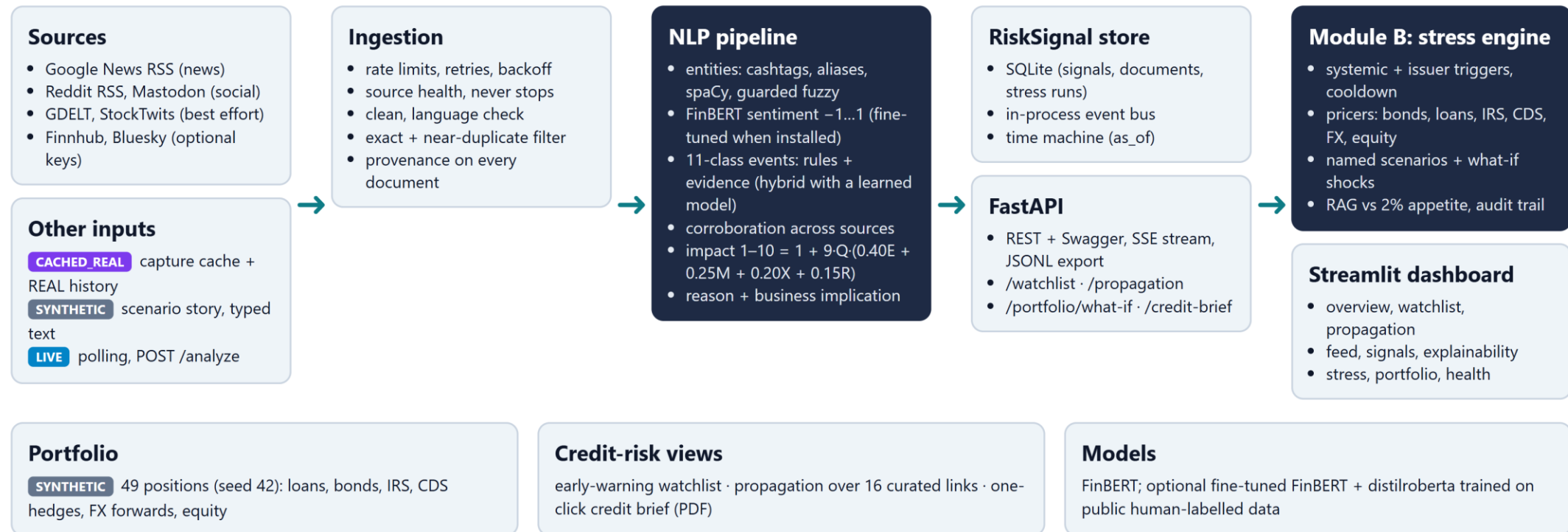
Measured: median 204.7 ms per document (n = 200)

System design

One NLP pipeline for live, replayed, scripted and API input

Risk Signal Engine: architecture

One NLP pipeline for live, replayed, scripted and API input · every record labelled LIVE / CACHED_REAL / SYNTHETIC



Stress results come from a simplified, illustrative hackathon model, not a production or regulatory risk model. Decision support, not investment advice.

Implementation highlights

Every score is explained, every weight is published

- **NLP:** entities, FinBERT sentiment, 11 event classes with evidence
- **Stress engine:** news triggers with cooldown; bonds, loans, swaps, CDS, FX, equity
- **Credit views:** watchlist, propagation, one-click credit brief
- **Engineering:** one pipeline for every input, offline demo, time machine, audit trail



Factor contributions for a SYNTHETIC demo signal

$$\text{Impact} = 1 + 9 \times Q \times (0.40 \cdot E + 0.25 \cdot M + 0.20 \cdot X + 0.15 \cdot R)$$

E severity · M sentiment · X exposure · R credibility · Q confidence · uncalibrated expert priors

Key results and metrics

Accuracy: preliminary (labels pending human review)

0.653

Sentiment

FinBERT · n = 147

0.81

Events

rules · n = 147

0.898

Entities

n = 147

204.7 ms

Median latency

n = 200 docs

84 → 50

Stress runs

n = 911 real docs

AI-drafted labels: a smoke test, not a benchmark.

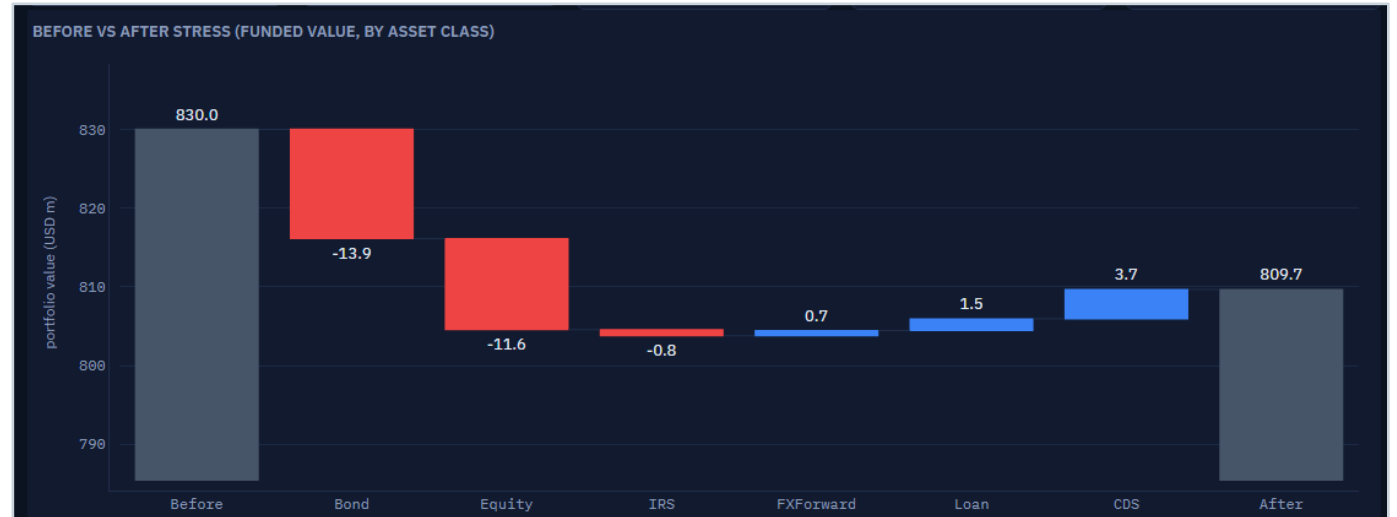
Scenario demo (simulated)

● **0.41% GREEN** Tata Motors downgrade

● **1.32% AMBER** Invasion (1 source)

● **2.45% RED** Corroborated (2 sources)

REAL history: 1,325 docs → 54 simulated runs



Before → after, corroborated geopolitical shock

Simplified, illustrative hackathon stress model. Not a production or regulatory risk model.

Simulated losses · illustrative model, synthetic portfolio · not investment advice

Domain impact and business value

Decision support for credit and portfolio risk

Credit analysts

see which held names need attention, and why

Risk managers

get an audited portfolio view when news breaks

Portfolio managers

see linked exposure and test their own shocks

WATCH-NEGATIVE 3 held issuers needing attention		MONITOR 6 held issuers with negative news		STABLE 14 no negative signal of note		SIGNALS IN WINDOW 51 on held issuers - last 24 h (event time)				
#	STATUS	ISSUER - RULE	RTG	EXPOSURE	PROPAG.	SIG.	WORST	EVENTS - SOURCES	24 H IMPACT	
1	WATCH	Tata Motors Ltd. (TATAMOTORS.NS) negative signal with impact 8.7 >= 7 (sentiment -0.90)	BB	\$41.8m 4.94%	\$23.4m	2 (1)	8.7	Earnings x1, Credit Event x1 google_news		Explain +
2	WATCH	JPMorgan Chase & Co. (JPM) negative signal with impact 7.7 >= 7 (sentiment -0.93)	A	\$43.4m 5.23%	\$34.9m	17 (7)	7.7	Other x13, Earnings x4 google_news, mastodon		Explain +
3	WATCH	Tesla Inc. (TSLA) 3 negative signals with impact >= 5	BBB	\$40.6m 4.89%	\$23.5m	12 (3)	6.7	Other x9, Earnings x3 google_news		Explain +
4	MONITOR	Microsoft Corp. (MSFT) negative signal with impact 5.2 >= 4	AAA-AA	\$24.8m 2.99%	\$18.6m	11 (1)	6.3	Other x9, Earnings x2 google_news		Explain +
5	MONITOR	Nvidia Corp. (NVDA) negative signal with impact 5.2 >= 4	AAA-AA	\$13.4m 1.62%	\$62.7m	5 (1)	6.2	Other x2, Supply Chain x1, Earnings x1, Product Launch x1 google_news, mastodon		Explain +
6	MONITOR	Bank of America Corp. 22 linked to US-JPM (peer_of, watched to us/JPM-AIGCATHF	A	\$40.3m 5.82%	\$22.0m	0 (0)	—	—	no signals	

Watchlist on real news

Credit brief - Tata Motors Ltd. (TATAMOTORS.NS)

Consumer Discretionary - IN - rating bucket BB (illustrative) - as of 2026-10-06T16:13 UTC - WATCH-NEGATIVE

Tata Motors Ltd. is WATCH-NEGATIVE on the early-warning watchlist (2 signals in the 24 h before 2026-10-06T16:13 UTC): negative signal with impact 8.7 >= 7 (sentiment -0.90).

We hold \$41.0m (4.94% of the funded book) across 3 positions, rating bucket BB. Linked issuers add \$23.4m of decay-weighted exposure (2.81%), 7.76% in total.

Under the issuer-only scenario "Idiosyncratic credit event (single issuer)", these positions would change by -\$5.3m, with CDS hedges offsetting \$1.9m (net -\$3.4m): 0.41% of the book.

In the latest systemic run (Geopolitical shock (severe), 2026-10-06T16:13 UTC) this issuer's positions moved -\$1.7m; the book lost 2.45% (RED).

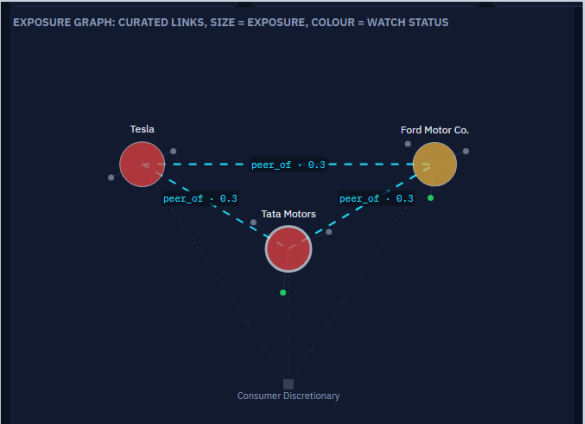
LINKED ISSUERS (CURATED)

US-TSLA (peer_of x0.3: \$12.2m), US-F (peer_of x0.3: \$11.2m)

LAST 3 SIGNALS

event time	impact	level	event	sent.	provenance	reason
2026-10-06T16:13	8.7	Critical	Credit Event	-0.90	SYNTHETIC	Impact 8.7 (Critical) driven mainly by Credit Event severity (E = 0.90) and strongly negative sentiment (s = -0.90).
2026-10-	5.6	Medium	Earnings	-0.06	CACHED_REAL	Impact 5.6 (Medium) driven mainly by Earnings severity (E = 0.55) and portfolio exposure to

Credit brief (PDF)



Propagation graph

No ROI claims: earlier attention, triage with the reason shown, audit trail to the headline

Limitations and future work

Limitations

- Uncalibrated impact weights
- Rules still misfire at times
- Headline-only news
- Small AI-labelled eval set
- Illustrative stress model

Future work

- Calibrate on market moves
- Licensed full-text news
- Fine-tuned models (ready)
- Streaming (Kafka)
- Backtesting

What already works

- Provenance on every record
- One pipeline for all input
- Real-data time machine
- Trigger audit trail
- Offline laptop demo

An honest, explainable bridge from text to portfolio risk, offline on a laptop